

Rohan Chandra

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 Publications on Google Scholar

(a) Professional Appointments

University of Virginia

Assistant Professor of Computer Science

Department of Computer Science, School of Engineering and Applied Science. Affiliated with the Link Lab. Founding member, UVA Physical AI Hub. Director, Chandra Robot Autonomy Lab.

Charlottesville, VA

June 2024 to present

University of Texas at Austin

Postdoctoral Research Fellow

Texas Robotics. Supervisors: Dr. Joydeep Biswas and Dr. Peter Stone. Research at the intersection of multi-agent systems, robotics, and control theory.

Austin, TX

June 2022 to June 2024

NVIDIA

Applied Research Intern (remote)

Autonomous driving prediction. Ego-vehicle trajectory and behavior prediction via ego-goal conditioning for hard maneuvers such as U-turns and unprotected left turns.

Santa Clara, CA

May 2021 to August 2021

University of Maryland, College Park

Graduate Teaching and Research Assistant

College Park, MD

August 2016 to May 2018

(b) Education and Postdoctoral Training

University of Texas at Austin

Postdoctoral Training, Texas Robotics

Advisors: Dr. Joydeep Biswas and Dr. Peter Stone.

Austin, TX

June 2022 to June 2024

University of Maryland, College Park

Ph.D. in Computer Science

Advisor: Dr. Dinesh Manocha. Dissertation: *Towards Autonomous Driving in Dense, Heterogeneous, and Unstructured Traffic*. Committee: Dr. Dinesh Manocha, Dr. Yiannis Aloimonos, Dr. Pratap Tokekar, Dr. Mac Schwager, Dr. Derek Paley.

College Park, MD

August 2018 to May 2022

University of Maryland, College Park

M.S. in Computer Science

College Park, MD

August 2016 to May 2018

Delhi Technological University

B.Tech. in Electronics and Communication Engineering

New Delhi, India

August 2012 to May 2016

(c) Honors and Awards

- (2026) 5th place worldwide, simulation round, **Benchmark Autonomous Robot Navigation (BARN) Challenge**, ICRA 2026. Qualified for the finals.
- (2026) **Washington Academy of Sciences Early Career Award** in Engineering and Computer Science.
- (2026) **AAAI 2026 New Faculty Highlights** selection.

- **(2026) Founding member, UVA Physical AI Hub.** One of 12 hubs selected by the Office of the Vice President for Research from over 90 proposals.
- **(2025) 3rd place worldwide, Benchmark Autonomous Robot Navigation (BARN) Challenge, ICRA 2025.**
- **(2025) NVIDIA Academic Grant Award.**
- **(2024) Drones 2023 Young Investigator Award, MDPI.**
- **(2023) Best Paper and Presentation Award, IROS 2023 Workshop on Advances in Multi-Agent Learning, for iPLAN.**
- **(2023) Seoul National University Ph.D. Talk Award, IROS 2023 Workshop on Integrated Perception, Planning, and Control.** Award amount \$1,000, sponsored by the SNU Research Center for Advanced Unmanned Vehicles.
- **(2023) Finalist, Charles A. Caramello Distinguished Dissertation Award,** a university level award at the University of Maryland, College Park across all graduate disciplines.
- **(2023) Selected as one of 10 speakers, Microsoft Future Leaders in Robotics and AI,** a joint University of Maryland and Microsoft initiative sponsored by the Microsoft DC Metro Engineering Site.
- **(2023) Selected (declined) for the Rising Stars in AI Symposium 2023 Speaker Series at KAUST.**
- **(2022) Finalist, UMD Innovation of the Year Award,** for work on deepfake detection.
- **(2022) Selected as one of 30 RSS Pioneers, Robotics: Science and Systems, New York.**
- **(2021) Finalist, UMD Innovation of the Year Award,** for work on multimodal and context aware emotion perception.
- **(2021) Selected as one of 30 participants, Future Faculty Fellow program, University of Maryland.**
- **(2020) Summer Research Fellowship, The Graduate School, University of Maryland, College Park.**

(d) Publications and Scholarly Work

Notation. Rohan Chandra denotes the candidate. Underlined names are student coauthors advised by the candidate. Journal impact factors are the most recent Clarivate Journal Citation Reports values available at the time of writing; the JCR year is stated with each figure. Page or article numbers follow the publisher of record. Conference acceptance rates are the figure published for that specific conference and year. Venues that publish no acceptance rate are listed without one.

d.1 Archival Peer Reviewed Journal Articles

1. Luc D'Costa, Yidi Wang, Jonathan L. Goodall, **Rohan Chandra**. "Physics-Informed CNN-LSTM for Street-Scale Urban Flood Prediction: Reconciling Aggregate Accuracy and Street-Level Plausibility." *Water (MDPI)*, vol. 18, no. 15, 2026, article 1809. *Impact factor*: 3.5.
2. Xuesu Xiao, Zifan Xu, Saad Abdul Ghani, Aniket Datar, Daeun Song, Peter Stone, Amna Mazen, Kamyab Yazdipaz, Innocent Mateyaunga, Mariam Faied, Mohan Krishnan, Yuanjie Lu, Tong Xu, Nick Mohammad, **Rohan Chandra**, Nicola Bezzo. "Autonomous Ground Navigation in Highly Constrained Spaces." *IEEE Robotics and Automation Magazine (RA-M)*, vol. 33, no. 1, 2026, pp. 186 to 196. *Impact factor*: 7.1 (JCR 2024).
3. **Rohan Chandra**, Shubham Singh, Wenhao Luo, Katia Sycara. "Multi-Robot Navigation in Social Mini-Games: Definitions, Taxonomy, and Algorithms." *Autonomous Robots (Springer)*, vol. 50, no. 2,

2026, article 20. *Impact factor*: 4.3 (JCR 2024).

4. **Rohan Chandra**, Vrushabh Zinage, Efstathios Bakolas, Peter Stone, Joydeep Biswas. "Deadlock-Free, Safe, and Decentralized Multi-Robot Navigation in Social Mini-Games via Discrete-Time Control Barrier Functions." *Autonomous Robots (Springer)*, vol. 49, no. 2, 2025, article 12. *Impact factor*: 4.3 (JCR 2024).
5. Anthony Francis, Claudia Perez-d'Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Rachid Alami, Aniket Bera, Abhijat Biswas, Joydeep Biswas, **Rohan Chandra**, Hao-Tien Lewis Chiang, Michael Everett, Sehoon Ha, Justin Hart, Jonathan P. How, Haresh Karnan, Tsang-Wei Edward Lee, Luis J. Manso, Reuth Mirsky, Sören Pirk, Phani Teja Singamaneni, Peter Stone, Ada V. Taylor, Peter Trautman, Nathan Tsoi, Marynel Vázquez, Xuesu Xiao, Peng Xu, Naoki Yokoyama, Alexander Toshev, Roberto Martín-Martín. "Principles and Guidelines for Evaluating Social Robot Navigation Algorithms." *ACM Transactions on Human-Robot Interaction (THRI)*, vol. 14, no. 2, 2025, pp. 1 to 65. *Impact factor*: 5.5 (JCR 2024).
6. Chen Tang, Ben Abbatematteo, Jiaheng Hu, **Rohan Chandra**, Roberto Martín-Martín, Peter Stone. "Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes." *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 8, no. 1, 2025, pp. 153 to 188. *Impact factor*: 14.0 (JCR 2024).
7. **Rohan Chandra**, Aniket Bera, Dinesh Manocha. "Using Graph-Theoretic Machine Learning to Predict Human Driver Behavior." *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, vol. 23, no. 3, March 2022 (early access 2021), pp. 2572 to 2585. *Impact factor*: 9.1 (JCR 2025).
8. **Rohan Chandra**, Ravi Teja, Neeta Pandey, Rajeshwari Pandey. "OTRA Based R-2R Ladder and Weighted Resistors DAC Realization." *International Journal of Electrical and Electronics Engineering*, vol. 7, no. 2, 2015, pp. 75 to 78.
9. Rashika Anurag, Neeta Pandey, **Rohan Chandra**, Rajeshwari Pandey. "Voltage Mode Second Order Notch and All-Pass Filter Realization Using OTRA." *i-Manager's Journal on Electronics Engineering*, vol. 6, no. 2, 2015, p. 22.
10. Arthur Benjamin, **Rohan Chandra**. "Multiplying by 9." *The College Mathematics Journal*, vol. 47, no. 4, 2016, p. 281.

d.2 Archival Peer Reviewed Journal Articles with Concurrent Conference Presentation (IEEE RA-L Dual Track).....

These are archival journal articles in *IEEE Robotics and Automation Letters* that were additionally presented at the indicated conference. They are separated from d.1 and d.3 so that the two categories are not double counted. RA-L impact factor: 5.3 (JCR 2024).

1. **Rohan Chandra**, Rahul Maligi, Arya Anantula, Joydeep Biswas. "SocialMAPF: Optimal and Efficient Multi-Agent Path Finding with Strategic Agents for Social Navigation." *IEEE Robotics and Automation Letters (RA-L)*, vol. 8, no. 6, 2023, pp. 3214 to 3221. *Impact factor*: 5.3 (JCR 2024). Presented at IROS 2023.
2. Tianrui Guan, Divya Kothandaraman, **Rohan Chandra**, Adarsh Jagan Sathyamoorthy, Kasun Weerakoon, Dinesh Manocha. "GA-Nav: Efficient Terrain Segmentation for Robot Navigation in Unstructured Outdoor Environments." *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 3, 2022, pp. 8138 to 8145. *Impact factor*: 5.3 (JCR 2024). Presented at IROS 2022.
3. Angelos Mavrogiannis, **Rohan Chandra**, Dinesh Manocha. "B-GAP: Behavior-Rich Simulation and Navigation for Autonomous Driving." *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2,

2022, pp. 4718 to 4725. *Impact factor*: 5.3 (JCR 2024). Presented at IROS 2022.

4. **Rohan Chandra**, Dinesh Manocha. "GamePlan: Game-Theoretic Multi-Agent Planning with Human Drivers at Intersections, Roundabouts, and Merging." *IEEE Robotics and Automation Letters (RA-L)*, vol. 7, no. 2, 2022, pp. 2676 to 2683. *Impact factor*: 5.3 (JCR 2024). Presented at ICRA 2022.
5. **Rohan Chandra**, Tianrui Guan, Srujan Panuganti, Trisha Mittal, Uttaran Bhattacharya, Aniket Bera, Dinesh Manocha. "Forecasting Trajectory and Behavior of Road-Agents Using Spectral Clustering in Graph-LSTMs." *IEEE Robotics and Automation Letters (RA-L)*, vol. 5, no. 3, 2020, pp. 4882 to 4890. *Impact factor*: 5.3 (JCR 2024). Presented at IROS 2020.

d.3 Archival Peer Reviewed Conference Proceedings.....

1. Wentao Zhou, Xuweiyi Chen, Vignesh Rajagopal, Jeffrey Chen, **Rohan Chandra**, Zezhou Cheng. "Empowering Dynamic Urban Navigation with Stereo and Mid-Level Vision." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, October 2026 (to appear). *Acceptance rate*: 36%.
2. Daeun Song, Nhat Le, Jeffrey Chen, Mohammad Nazeri, Amirreza Payandeh, **Rohan Chandra**, Reuth Mirsky, Ross Mead, Ling Xiao, Xuesu Xiao. "HuMAIN: Human-Aware Implicit Social Robot Navigation." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, October 2026 (to appear). *Acceptance rate*: 36%.
3. Amir Moeini, Minjae Kwon, Alper Kamil Bozkurt, Yuichi Motai, **Rohan Chandra**, Lu Feng, Shangdong Zhang. "Safe In-Context Reinforcement Learning." *International Conference on Machine Learning (ICML)*, July 2026. *Acceptance rate*: 26.6%.
4. Timur Anvar, Jeffrey Chen, Yuyan Wang, **Rohan Chandra**. "Are LLMs The Way Forward? A Case Study on LLM-Guided Reinforcement Learning for Decentralized Autonomous Driving." *IEEE International Conference on Robotics and Automation (ICRA)*, June 2026. *Acceptance rate*: 38.04%.
5. Kefan Song, Amir Moeini, Peng Wang, Lei Gong, **Rohan Chandra**, Shangdong Zhang, Yanjun Qi. "Reward Is Enough: LLMs Are In-Context Reinforcement Learners." *International Conference on Learning Representations (ICLR)*, April 2026, pp. 112747 to 112770. *Acceptance rate*: 28%.
6. Xijun Wang, Pedro Sandoval-Segura, Chengyuan Zhang, Junyun Huang, Tianrui Guan, Ruiqi Xian, Fuxiao Liu, **Rohan Chandra**, Boqing Gong, Dinesh Manocha. "DAVE: Diverse Atomic Visual Elements Dataset with High Representation of Vulnerable Road Users in Complex and Unpredictable Environments." *IEEE Intelligent Vehicles Symposium (IV)*, June 2026, pp. 779 to 786.
7. Zixuan Xie, Xinyu Liu, **Rohan Chandra**, Shangdong Zhang. "Finite Sample Analysis of Linear Temporal Difference Learning with Arbitrary Features." *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 38, December 2025, pp. 96025 to 96057. *Acceptance rate*: 24.52%.
8. Jiuqi Wang, **Rohan Chandra**, Shangdong Zhang. "Towards Provable Emergence of In-Context Reinforcement Learning." *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 38, December 2025, pp. 25873 to 25908. *Acceptance rate*: 24.52%.
9. Vagul Mahadevan, Shangdong Zhang, **Rohan Chandra**. "GameChat: Multi-LLM Dialogue for Safe, Agile, and Socially Optimal Multi-Agent Navigation in Constrained Environments." *IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, 2025, pp. 1 to 7. *Acceptance rate*: 47.8%.
10. Nilesh Suriyarachchi, Arya Anantula, John S. Baras, Dinesh Manocha, **Rohan Chandra**. "Efficient Realtime Traffic Routing for Heterogeneous Agents in Unregulated Intersections." *IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, 2025, pp. 1 to 7. *Acceptance rate*: 47.8%.

Circulated in preprint as GAMEOPT+.

11. **Rohan Chandra**, Haresh Karnan, Negar Mehr, Peter Stone, Joydeep Biswas. "Multi-Agent Inverse Reinforcement Learning in Real World Unstructured Pedestrian Crowds." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, October 2025, pp. 18668 to 18675. *Acceptance rate*: 46.2%.
12. Kefan Song, Jin Yao, Runnan Jiang, **Rohan Chandra**, Shangdong Zhang. "Towards Large Language Models That Benefit for All: Benchmarking Group Fairness in Reward Models." *Reinforcement Learning Conference (RLC)*, August 2025. *Acceptance rate*: 39%.
13. Srikar Gouru, Siddharth Lakkoju, **Rohan Chandra**. "LiveNet: Robust, Minimally Invasive Multi-Robot Control for Safe and Live Navigation in Constrained Environments." *Annual Learning for Dynamics and Control Conference (L4DC)*, June 2025.
14. Vrushabh Zinage, Abhishek Jha, **Rohan Chandra**, Efstathios Bakolas. "Decentralized Safe and Scalable Multi-Agent Control under Limited Actuation." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2025, pp. 12105 to 12111. *Acceptance rate*: 38%.
15. Jennifer Williams, Eike Schneiders, Henry Card, Tina Seabrooke, Beatrice Pakenham-Walsh, Tayyaba Azim, Lucy Valls-Reed, Ganesh Vigneswaran, John Robert Bautista, **Rohan Chandra**, Arya Farahi. "Predicting Acute Pain Levels Implicitly from Vocal Features." *INTERSPEECH*, September 2024, pp. 1460 to 1464. *Acceptance rate*: approximately 50%.
16. Vrushabh Zinage, **Rohan Chandra**, Efstathios Bakolas. "Disturbance Observer-Based Robust Integral Control Barrier Functions for Nonlinear Systems with High Relative Degree." *American Control Conference (ACC)*, July 2024, pp. 2470 to 2475. **Oral presentation**.
17. Amir Hossain Raj, Zichao Hu, Haresh Karnan, **Rohan Chandra**, Amirreza Payandeh, Luisa Mao, Peter Stone, Joydeep Biswas, Xuesu Xiao. "Rethinking Social Robot Navigation: Leveraging the Best of Two Worlds." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2024, pp. 16330 to 16337. *Acceptance rate*: 44%.
18. Xiyang Wu, **Rohan Chandra**, Tianrui Guan, Amrit Singh Bedi, Dinesh Manocha. "Intent-Aware Planning in Heterogeneous Traffic via Distributed Multi-Agent Reinforcement Learning." *Conference on Robot Learning (CoRL)*, PMLR, November 2023, pp. 446 to 477. *Acceptance rate*: 40.0%. **Oral presentation**. Circulated as iPLAN.
19. **Rohan Chandra**, Xijun Wang, Mridul Mahajan, Rahul Kala, Rishitha Palugulla, Chandrababu Naidu, Alok Jain, Dinesh Manocha. "METEOR: A Massive Dense and Heterogeneous Behavior Dataset for Autonomous Driving." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2023, pp. 9169 to 9175. *Acceptance rate*: 45%.
20. Nilesh Suriyarachchi, **Rohan Chandra**, John S. Baras, Dinesh Manocha. "GAMEOPT: Optimal Real-Time Multi-Agent Planning and Control at Dynamic Intersections." *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, October 2022, pp. 2599 to 2606. *Acceptance rate*: approximately 50.9%.
21. **Rohan Chandra**, Mingyu Wang, Mac Schwager, Dinesh Manocha. "Game-Theoretic Planning for Autonomous Driving among Risk-Aware Human Drivers." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2022, pp. 2876 to 2883. *Acceptance rate*: 45%.
22. Tianrui Guan, Jun Wang, Shiyi Lan, **Rohan Chandra**, Zuxuan Wu, Larry Davis, Dinesh Manocha. "M3DeTR: Multi-Representation, Multi-Scale, Mutual-Relation 3D Object Detection with Transformers." *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, January 2022, pp. 2293

to 2303. *Acceptance rate: 35.0%.*

23. Trisha Mittal, Uttaran Bhattacharya, **Rohan Chandra**, Aniket Bera, Dinesh Manocha. "Emotions Don't Lie: An Audio-Visual Deepfake Detection Method Using Affective Cues." *ACM International Conference on Multimedia (ACM MM)*, October 2020, pp. 2823 to 2832. *Acceptance rate: 27.9%.*
24. **Rohan Chandra**, Uttaran Bhattacharya, Trisha Mittal, Aniket Bera, Dinesh Manocha. "CMetric: A Driving Behavior Measure Using Centrality Functions." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, October 2020, pp. 2035 to 2042. *Acceptance rate: 45%.*
25. Uttaran Bhattacharya, Christian Roncal, Trisha Mittal, **Rohan Chandra**, Kyra Kapsaskis, Kurt Gray, Aniket Bera, Dinesh Manocha. "Take an Emotion Walk: Perceiving Emotions from Gaits Using Hierarchical Attention Pooling and Affective Mapping." *European Conference on Computer Vision (ECCV)*, Springer, August 2020, pp. 145 to 163. *Acceptance rate: 26.4%.*
26. Adarsh Jagan Sathyamoorthy, Jing Liang, Utsav Patel, Tianrui Guan, **Rohan Chandra**, Dinesh Manocha. "DenseCAvoid: Real-Time Navigation in Dense Crowds Using Anticipatory Behaviors." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2020, pp. 11345 to 11352. *Acceptance rate: 40%.*
27. Trisha Mittal, Pooja Guhan, Uttaran Bhattacharya, **Rohan Chandra**, Aniket Bera, Dinesh Manocha. "EmotiCon: Context-Aware Multimodal Emotion Recognition Using Frege's Principle." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020, pp. 14234 to 14243. *Acceptance rate: 22.1%.*
28. **Rohan Chandra**, Uttaran Bhattacharya, Trisha Mittal, Xiaoyu Li, Aniket Bera, Dinesh Manocha. "GraphRQI: Classifying Driver Behaviors Using Graph Spectrums." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2020, pp. 4350 to 4357. *Acceptance rate: 40%.*
29. **Rohan Chandra**, Uttaran Bhattacharya, Trisha Mittal, Aniket Bera, Dinesh Manocha. "RoadTrack: Tracking Road Agents in Dense and Heterogeneous Environments." *IEEE International Conference on Robotics and Automation (ICRA)*, May 2020, pp. 1270 to 1277. *Acceptance rate: 40%.*
30. Uttaran Bhattacharya, Trisha Mittal, **Rohan Chandra**, Tanmay Randhavane, Aniket Bera, Dinesh Manocha. "STEP: Spatial Temporal Graph Convolutional Networks for Emotion Perception from Gaits." *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 34, no. 2, February 2020, pp. 1342 to 1350. *Acceptance rate: 20.6%.*
31. Trisha Mittal, Uttaran Bhattacharya, **Rohan Chandra**, Aniket Bera, Dinesh Manocha. "M3ER: Multiplicative Multimodal Emotion Recognition Using Facial, Textual, and Speech Cues." *AAAI Conference on Artificial Intelligence (AAAI)*, vol. 34, no. 2, February 2020, pp. 1359 to 1367. *Acceptance rate: 20.6%. Oral presentation.*
32. **Rohan Chandra**, Uttaran Bhattacharya, Christian Roncal, Aniket Bera, Dinesh Manocha. "RobustTP: End-to-End Trajectory Prediction for Heterogeneous Road-Agents in Dense Traffic with Noisy Sensor Inputs." *ACM Computer Science in Cars Symposium (ACM CSCS)*, October 2019, pp. 1 to 9.
33. **Rohan Chandra**, Tom Goldstein, Christoph Studer. "PhasePack: A Phase Retrieval Library." *International Conference on Sampling Theory and Applications (SampTA)*, IEEE, 2019, pp. 1 to 5.
34. **Rohan Chandra**, Uttaran Bhattacharya, Aniket Bera, Dinesh Manocha. "DensePeds: Pedestrian Tracking in Dense Crowds Using Front-RVO and Sparse Features." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, November 2019, pp. 468 to 475. *Acceptance rate: 45%.*
35. **Rohan Chandra**, Uttaran Bhattacharya, Aniket Bera, Dinesh Manocha. "TraPHic: Predicting Trajectories of Road-Agents in Dense and Heterogeneous Traffic." *IEEE/CVF Conference on Computer*

Vision and Pattern Recognition (CVPR), June 2019, pp. 8475 to 8484. *Acceptance rate*: 25.2%.

d.4 Books.....

None to date.

d.5 Book Chapters.....

Rohan Chandra, Dinesh Manocha. "Perception for Behavior Modeling and Classification in ADAS." In S. Shekhar, H. Xiong, X. Zhou (eds.), *Encyclopedia of GIS*, Springer, Cham, 2026. DOI 10.1007/978-3-319-23519-6_1667-1.

d.6 Peer Reviewed Workshop Proceedings.....

Listed separately from d.3 because these appeared in the ICCV Workshops (ICCVW) proceedings rather than the ICCV main conference.

1. Divya Kothandaraman, **Rohan Chandra**, Dinesh Manocha. "BoMuDANet: Unsupervised Adaptation for Visual Scene Understanding in Unstructured Driving Environments." *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, October 2021, pp. 3949 to 3958.
2. Divya Kothandaraman, **Rohan Chandra**, Dinesh Manocha. "SS-SFDA: Self-Supervised Source-Free Domain Adaptation for Road Segmentation in Hazardous Environments." *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, October 2021, pp. 3042 to 3052.

d.7 Other Peer Reviewed Scholarly Work (Demonstration and Highlight Tracks).....

Separated from d.3 because these appeared in short non-archival-research tracks. Both carry page numbers in the AAAI proceedings of record.

1. **Rohan Chandra**. "Towards Agents That Exhibit Human-Like Autonomy in Complex Environments." *AAAI Conference on Artificial Intelligence (AAAI), New Faculty Highlights Track*, vol. 40, no. 47, February 2026, pp. 39809 to 39810. Invited, competitively selected track.
2. **Rohan Chandra**, Zayne Sprague, Joydeep Biswas. "SocialGym 2.0: Simulator for Multi-Agent Social Robot Navigation in Shared Human Spaces." *AAAI Conference on Artificial Intelligence (AAAI), Demonstration Track*, vol. 38, no. 21, February 2024, pp. 23778 to 23780. *Acceptance rate*: 23.75%.

d.8 Preprints and Manuscripts Under Review.....

Preprint versions of work already published in d.1 through d.6 are excluded. Venue and current review status should be added for each entry before submission.

1. Jeffrey Chen, **Rohan Chandra**. "Dynamic Control Barrier Function Regulation with Vision-Language Models for Safe, Adaptive, and Realtime Visual Navigation." arXiv:2603.21142, 2026. In preparation for submission to ICRA 2027.
2. Zixuan Xie, Xinyu Liu, **Rohan Chandra**, Shangdong Zhang. "Convergence and Emergence of In-Context Reinforcement Learning with Chain of Thought." arXiv:2605.07123, 2026. In preparation for submission.
3. Zixuan Xie, Xinyu Liu, Claire Chen, Shuze Daniel Liu, **Rohan Chandra**, Shangdong Zhang. "Beyond Linear Attention: Softmax Transformers Implement In-Context Reinforcement Learning." arXiv:2605.07333, 2026. In preparation for submission.
4. Jaskirat Singh, **Rohan Chandra**. "FACA: Fair and Agile Multi-Robot Collision Avoidance in Constrained Environments with Dynamic Priorities." arXiv:2511.14024, 2025. In preparation for submission.

5. Aleesha Khurram, Amir Moeini, Shangdong Zhang, **Rohan Chandra**. "Prompt-Driven Domain Adaptation for End-to-End Autonomous Driving via In-Context RL." arXiv:2511.12755, 2025. Work being extended by another student and in preparation for resubmission to ICRA 2027.
6. Vignesh Rajagopal, Kasun Weerakoon, Gershon Devake Seneviratne, Pon Aswin Sankaralingam, Mohamed Elnoor, Jing Liang, **Rohan Chandra**, Dinesh Manocha. "DR. Nav: Semantic-Geometric Representations for Proactive Dead-End Recovery and Navigation." arXiv:2511.12778, 2025. In preparation for resubmission.
7. Jeffrey Chen, **Rohan Chandra**. "LivePoint: Fully Decentralized, Safe, Deadlock-Free Multi-Robot Control in Cluttered Environments with High-Dimensional Inputs." arXiv:2503.13098, 2025. In preparation for submission to RA-L 2026.
8. Jiuqi Wang, **Rohan Chandra**, Shangdong Zhang. "Experience Replay Addresses Loss of Plasticity in Continual Learning." arXiv:2503.20018, 2025. In preparation for re-submission.
9. Kefan Song, Runnan Jiang, **Rohan Chandra**, Shangdong Zhang. "Group Fairness in Multi-Task Reinforcement Learning." arXiv:2503.07817, 2025. In preparation for re-submission.
10. Amir Moeini, Jiuqi Wang, Jacob Beck, Ethan Blaser, Shimon Whiteson, **Rohan Chandra**, Shangdong Zhang. "A Survey of In-Context Reinforcement Learning." arXiv:2502.07978, 2025. In preparation for re-submission.
11. Chirag Parikh, Ravi Shankar Mishra, **Rohan Chandra**, Ravi Kiran Sarvadevabhatla. "Transfer-LMR: Heavy-Tail Driving Behavior Recognition in Diverse Traffic Scenarios." arXiv:2405.05354, 2024. In preparation for re-submission.
12. Vrushabh Zinage, **Rohan Chandra**, Efstathios Bakolas. "Neural Differentiable Integral Control Barrier Functions for Unknown Nonlinear Systems with Input Constraints." arXiv:2312.07345, 2023. In preparation for re-submission.
13. **Rohan Chandra**, Rahul Menon, Zayne Sprague, Arya Anantula, Joydeep Biswas. "Decentralized Social Navigation with Non-Cooperative Robots via Bi-Level Optimization." arXiv:2306.08815, 2023.
14. Trisha Mittal, Puneet Mathur, **Rohan Chandra**, Apurva Bhatt, Vikram Gupta, Debdoot Mukherjee, Aniket Bera, Dinesh Manocha. "Estimating Emotion Contagion on Social Media via Localized Diffusion in Dynamic Graphs." arXiv:2207.07165, 2022.
15. **Rohan Chandra**, Sachin Grover, Kyungjun Lee, Moustafa Meshry, Ahmed Taha. "Texture Synthesis with Recurrent Variational Auto-Encoder." arXiv:1712.08838, 2017.

(e) Citation Metrics

Citation organization: Google Scholar (profile), retrieved August 22, 2026.

- Total citations: 4951
- h-index: 29
- i10-index: 47
- Citations since 2021: 4689

(f) Graduate Students Directed

f.1 Doctoral Students Directed

Aneesh Chavan, Ph.D. in Computer Science, started August 2025. Status: 2nd year, coursework completed. Most recent milestone: completed rotation. Projected defense: 2030.

Vignesh Rajagopal, Ph.D. in Computer Science, started August 2025. Status: 2nd year, coursework completed. Most recent milestone: completed rotation. Projected defense: 2030. First author on one manuscript under review (DR. Nav).

Jeffrey Chen, Ph.D. in Computer Science, started August 2024. Status: 2nd year, coursework completed. Most recent milestone: qualifying proposal passed. Projected defense: 2029. Coauthor on two IROS 2026 papers and one ICRA 2026 paper; first author on two manuscripts under review.

f.2 Master of Science Students Directed.....

Donggen Li, M.S. in Computer Science, started April 2025. Status: graduated. Most recent milestone: M.S. completed. Developed the algorithm for the BARN Challenge entry.

f.3 Master of Engineering Students Directed.....

None to date.

Committee service for students not directed by the candidate is departmental service and is listed under (I.1).

(g) Undergraduate Student Researchers

- **Azim Abdulmajeeth** (2026 to present).
- **Claudia Castagna** (2026 to present).
- **Luc D'Costa** (2026 to present). First author on one journal article in *Water* (MDPI), 2026.
- **Max Black** (2026 to present).
- **Sean Ruan** (2026 to present).
- **Himesh Ahuja** (2025 to present). Recipient of a \$2,000 Ingrassia Family Research Grant.
- **Srikar Gouri** (2025). First position: M.S., Carnegie Mellon University. First author on one L4DC 2025 paper (LiveNet).
- **Vagul Mahadevan** (2025). First position: Metron. First author on one MRS 2025 paper (GameChat).

- **Aleesha Khurram** (2025). First position: M.S., University of Oxford. First author on one manuscript under review.
- **Timur Anvar** (2025). First author on one ICRA 2026 paper.
- **Yuyan Wang** (2025). First position: M.S., University of Pennsylvania. Coauthor on one ICRA 2026 paper.
- **Siddharth Lakkoju** (2024). First position: SpaceX. Coauthor on one L4DC 2025 paper (LiveNet).
- **Rahul Maligi** (2023). First position: Amazon. Coauthor on one RA-L 2023 paper (SocialMAPF).
- **Arya Anantula** (2023). First position: M.S., Georgia Tech. Coauthor on one RA-L 2023 paper (SocialMAPF) and one MRS 2025 paper.

(h) Visitors and Postdoctoral Fellows Supervised

None to date.

(j) Presentations

j.1 Keynote Presentations.....

“Human-like Mobility to Deploy Autonomous Vehicles Everywhere.” Keynote, Workshop on Scenario and Behavior Diversity in Simulation for Autonomous Vehicle Validation. Inviting organization: IEEE International Automated Vehicle Validation Conference (IAVVC) 2023 workshop organizing committee. October 2023.

j.2 Invited Presentations.....

“Towards Agents That Exhibit Human-Like Autonomy in Complex Environments.” New Faculty Highlights invited talk. Inviting organization: Association for the Advancement of Artificial Intelligence (AAAI). January 2026.

“Autonomous Driving in Unstructured Traffic Environments.” First Workshop on Intelligent Mobility in Unstructured Environments. Inviting organization: International Conference on Pattern Recognition (ICPR) 2024 workshop organizing committee. December 2024.

“Human-like Mobility to Deploy Autonomous Vehicles Everywhere.” Symposium on Indian Driving Simulation. Inviting organization: Robotics Research Center, IIIT Hyderabad, India. December 2023.

“SocialGym 2.0: Simulator for Multi-Agent Social Robot Navigation in Shared Human Spaces.” 2nd Workshop on Social Robot Navigation: Advances and Evaluation. Inviting organization: IROS 2023 workshop organizing committee. October 2023.

“Human-like Mobility to Deploy Robots Everywhere.” Workshop on Integrated Perception, Planning, and Control for Physically and Contextually-Aware Robot Autonomy. Inviting organization: IROS 2023 workshop organizing committee. October 2023.

“Autonomous Driving in Unstructured Traffic Environments.” Department seminar. Inviting organization: IIIT Hyderabad, India. November 2022.

“Autonomous Driving in Unstructured Traffic Environments.” Department seminar. Inviting organization: IIIT Delhi, India. November 2022.

“Autonomous Driving in Unstructured Traffic Environments.” Department seminar. Inviting organization: Department of Computer Science and Engineering, IIT Delhi, India. November 2022.

“Autonomous Driving in Unstructured Traffic Environments.” Hazard Perception in Intelligent Vehicles (HPIV) Workshop. Inviting organization: WACV 2022 workshop organizing committee. January 2022.

“Human Driver Behavior Modeling in Dense Urban Traffic with Applications to Planning and Navigation.” Workshop on Perception and Control for Autonomous Navigation in Crowded, Dynamic Environments. Inviting organization: Robotics: Science and Systems (RSS) 2021 workshop organizing committee. July 2021.

“Modeling Human Driver Behavior in Dense Urban Traffic Environments Using Graph Theory.” Student Seminar Series. Inviting organization: Maryland Robotics Center, University of Maryland, College Park. March 2021.

(k) Patents, Disclosures, and Copyrights

k.1 Issued Patents

1. Trisha Mittal, Uttaran Bhattacharya, **Rohan Chandra**, Aniket Bera, Dinesh Manocha. "System and Method for Detecting Fabricated Videos." **US Patent 12,322,175 B2**, issued [date]. Filed as application 17/515,846, 2021.
2. Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, **Rohan Chandra**, Pooja Guhan, Dinesh Manocha. "Human Emotion Recognition in Images or Video." **US Patent 11,861,940 B2**, issued [date]. Filed as application 17/349,732, 2021.
3. Trisha Mittal, Aniket Bera, Uttaran Bhattacharya, **Rohan Chandra**, Dinesh Manocha. "System and Method for Multimodal Emotion Recognition." **US Patent 11,830,291 B2**, issued [date]. Filed as application 17/173,018, 2021. Supported by Army Research Office award W911NF1810313.

k.2 Filed Patent Applications Pending

None. All three applications have issued and are listed under k.1.

k.3 Invention Disclosures

None.

k.4 Formal Copyrights and Released Research Software

No formal copyright registrations. The research software below is released publicly under open source licenses, which is a licensing choice rather than a registration.

- o **Rohan Chandra**. Behavior-Driven-Traffic-Simulator. Released research software, 2019.
- o **Rohan Chandra**, Zayne Sprague, Joydeep Biswas. SocialGym 2.0 simulator. Released research software, 2024. MIT License.
- o Srikar Gouri, Siddharth Lakkoju, **Rohan Chandra**. LiveNet. Released research software, 2025. MIT License.

(l) Internal Service and Leadership Since Joining UVA

l.1 Department

- o Industry Engagement Committee (2026 to 2027).
- o Industry Engagement Committee (2025 to 2026).
- o Tenured and Tenure Track Faculty Search Committee, Department of Computer Science (2024 to 2025).
- o **Doctoral and master's dissertation committee service** (students not directed by the candidate), most recent first:
 - **2026:** Jinhao Liang (Ph.D.), Malori Oxford (Ph.D.), Guangyi Xu (M.S.).
 - **2025:** Shaid Hassan (Ph.D.), Jingyun Ning (Ph.D.), Nicholas Mohammad (Ph.D.), Jonathan Reasoner (Ph.D.), Wentao Zhou (M.S.), Amir Moeini (M.S.).

l.2 School

- o **Organizer and Chair**, first **Robotics + X Symposium** at UVA (2026), in collaboration with the Office of the Vice President for Research, funded by a SEAS Research Initiation Grant with co-organizers Yen-Ling Kuo and Tariq Iqbal.
- o Panelist, AI Panel hosted by the Link Lab (Spring 2025).
- o Link Lab Research Day laboratory tour host.

I.3 University.....

- **Founding member**, UVA **Physical AI Hub** (2026), one of 12 hubs selected by the Office of the Vice President for Research from over 90 proposals.
- Robotics demonstration (humanoid robot, BARN Challenge platform, and autonomous racing) for the Goochland Middle School Career and Technical Education class, 7th grade. 2026.

(m) Professional Service

m.1 Editorial Board Positions (Leadership Roles).....

- **Associate Editor**, *IEEE International Conference on Robotics & Automation* (ICRA), 2027.
- **Associate Editor**, *IEEE Robotics and Automation Letters* (RA-L), 2023 to present.
- **Associate Editor**, *IEEE Transactions on Robotics* (T-RO). Invited, declined, 2026.
- **Field Editor**, *Encyclopedia of GIS*, Springer, August 2026.
- **Guest Editor**, *MDPI Drones*, Special Issue on Advances in Perception and Artificial Intelligence for Autonomous Vehicles, 2022 to 2024.

m.2 Conference Leadership and Organizer Roles.....

- **Area Chair**, AAAI 2027.
- **Area Chair**, CoRL 2026.
- **Area Chair**, AAAI 2026.
- **Area Chair**, RLC 2025.
- **Organizer**, Workshop on Multi-Agent Planning and Navigation in Challenging Environments, RSS 2023.
- **Organizer**, 2nd Workshop on Social Robot Navigation: Advances and Evaluation, IROS 2023.
- **Organizer**, Workshop on Behavior-driven Autonomous Driving in Unstructured Environments, IROS 2022.
- **Publicity and Social Media Chair**, International Symposium on Multi-Robot and Multi-Agent Systems (MRS) 2023.
- **Session Chair**, Computer Vision for Autonomous Driving session, AAAI 2023.
- **Session Co-Chair**, Intelligent Transportation session, ICRA 2022.
- Program committee, ICCV 2021 Workshop on Multi-Agent Interaction and Relational Reasoning.

m.3 Grant Panel Service.....

- Panel reviewer, National Science Foundation (2026).
- Panel reviewer, National Science Foundation (2024).

m.4 Mentoring and Community Service to the Profession.....

- **Faculty mentor**, WACV 2026 Doctoral Consortium.
- Judge, special awards category for AAAI, Regeneron International Science and Engineering Fair (ISEF) 2023.
- Speaker, Future Leaders in Robotics and AI Seminar Series, University of Maryland and Microsoft Robotics and Diversity Initiative (2023).

m.5 Reviewing.....

- Reviewer for leading robotics, vision, and AI venues, 2018 to present: CVIU, IJCAI, CoRL, CVPR, AAAI, ICRA, IROS, RA-L, NeurIPS, ICLR, ICML, ICCV, RSS.

m.6 Prior Institutional Service.....

- Graduate Admissions Committee, Department of Computer Science, University of Texas at Austin (2022 to 2023).

- Graduate Admissions Committee, Department of Computer Science, University of Maryland, College Park (2017 to 2018).

Supplementary: Courses Taught at UVA

Provided for reference only. The full course table with enrollments and evaluation scores appears in the Teaching Portfolio (dossier item 5a).

CS 2120: Discrete Mathematics and Theory I **University of Virginia**
Fall 2026

CS 6501: Multi-Robot Navigation **University of Virginia**
Spring 2026

CS 2120: Discrete Mathematics and Theory I **University of Virginia**
Fall 2025

CS 6501: Multi-Robot Navigation **University of Virginia**
Spring 2025

Two class projects resulted in peer reviewed conference publications.

CS 2120: Discrete Mathematics and Theory I **University of Virginia**
Fall 2024

Supplementary: Outreach Prior to UVA

- Demonstrated Texas Robotics autonomous mobile robot navigation to 20 high school and middle school students, Robotics Summer Camp 2023, UT Austin.
- Volunteer, STEM Girl Day, UT Austin (2023).
- AI4ALL 2021: led a two week machine learning project for high school students.
- NYU AI School 2021: taught introductory machine learning and programming to students from underrepresented groups.
- AI4ALL 2020: taught introductory machine learning and programming to students from underrepresented groups.